Ch 6.4: Curse of Dimensionality

Lecture 17 - CMSE 381

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Announcements

Last time:

- PCA/PCR
- PLS

This lecture:

- Issues with higher dimensions

Midterm

Announcements:

- NO Homework due Friday
- Friday lab

Section 1

Last time

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

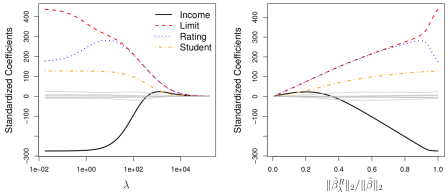
Least Squares:
No constraints

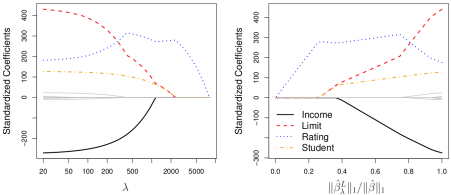
Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$





Linear transformation of predictors

Original Predictors:

big p

$$X_1, \cdots, X_p$$

New Predictors:

$M \ll P$

$$Z_1, \cdots, \cancel{X}_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

coeff

The goal:

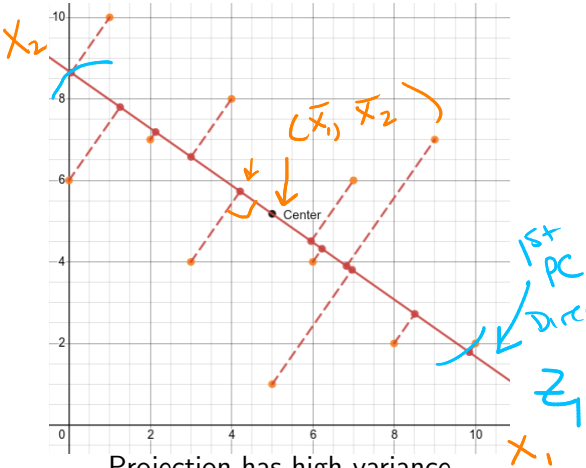
- Find good φ's
- Fit regression model on Z_i's using least squares

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

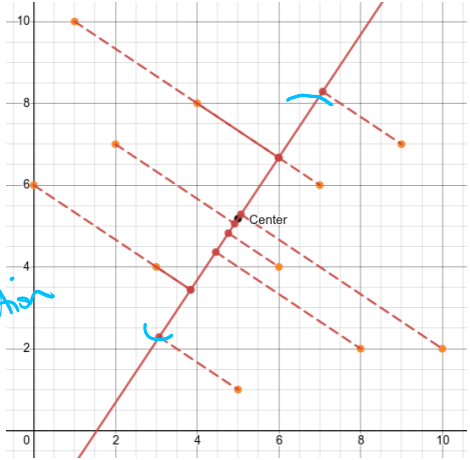
- Hope that lower dimensions means less overfitting
- Remember that interpretation not the same as shrinkage/subset selection of variables

Z_i

PCA - First PC

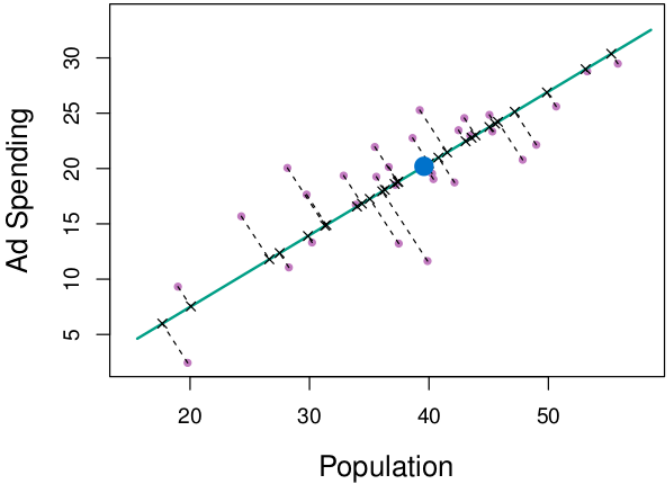


Projection has high variance



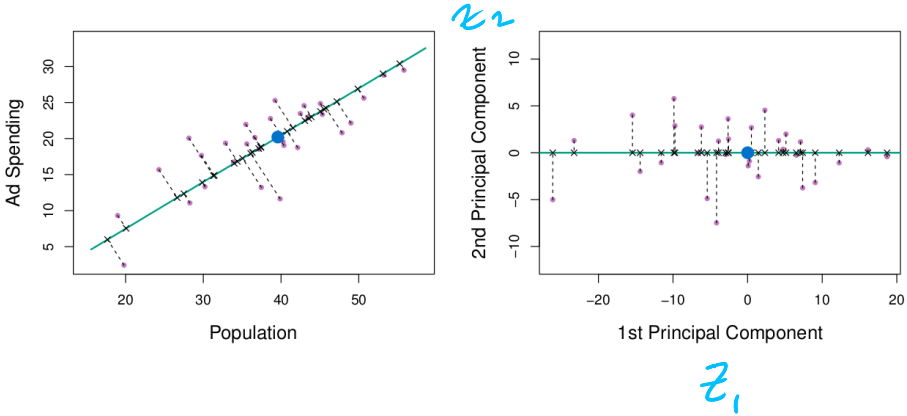
Projection has low(er) variance

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathtt{pop} - \overline{\mathtt{pop}}) + 0.544 \cdot (\mathtt{ad} - \overline{\mathtt{ad}})$$

Drawing points in PC space



Principal Components Regression (PCR)

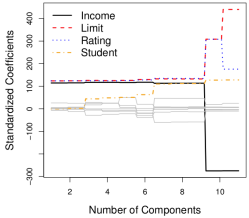
$M \leq p$

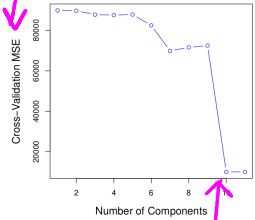
$\boxed{Z_1, Z_2, Z_3, \ldots}$ $\ldots, Z_p$

stop @ Z_m

approximates test MSE

- Take new features Z_i
- Run regression
- Maybe do CV for a bunch of choices of M (where M = number of Z_i features used) to pick a best M





$p = 11$

huge drop @ $M=10$

First direction Z_1 for Partial Least Squares (PLS)

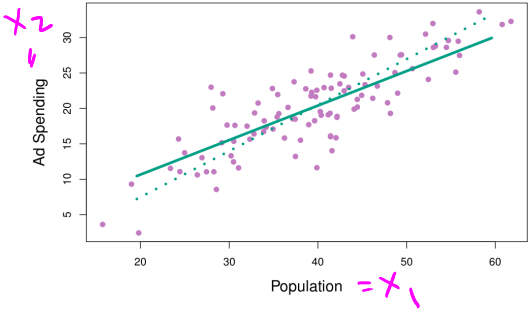
- Set φ_{j1} equal to the coefficient from simple linear regression of Y onto X_j
- The first direction is

$$Z_1 = \sum_{j=1}^{p} \varphi_{j1} X_j$$

$X_1, X_2, \ldots, X_p$

X_1 vs Y φ_{j1}

X_2 vs Y φ_{j2}

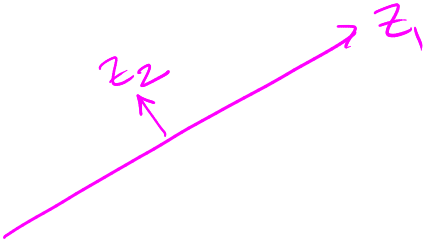


Ex. Prediction of Y =Sales (not shown) on X_1 =Population and X_2 =Ad Spending

- Solid green: First PLS direction
- Dashed: First PC direction

Second (and more) PLS directions

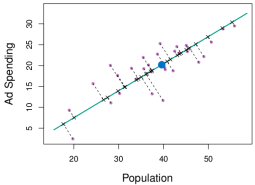
- Regress each variable on Z_1 and take residuals
- Compute Z_2 using *orthogonalized data* same as for Z_1



TL;DR

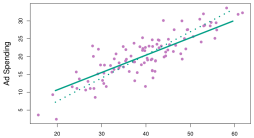
PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PLS

- Supervised dimensionality reduction
- Choose component Z_1 by using simple regression coefficients of each X_i onto Y
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained
- Not a particular benefit, so usually default to PCA unless you have a good reason for this



Section 2

Issues in Higher Dimensions

High-Dimensional Data

Low-Dimensions

$n \gg p$

↳ Low # X_i features relative to # data pts n

"Everything" prev in class works well here

Historically

Predict blood pressure using age, sex, BMI

$p=3$

High-Dimensions

$n \ll p$, $p \approx n$

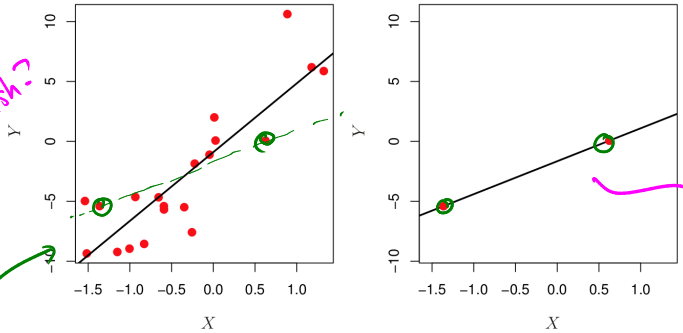
Predict blood pressure given genetic data

Classical approaches not appropriate

↳ overfitting

↳ Bias/Variance trade off

What goes wrong?



$p > n$

perfect fit
||
overfitting

$n = 20$, ish?

$n = 2$

perfect fit

not a perfect fit
||
(overfitting)

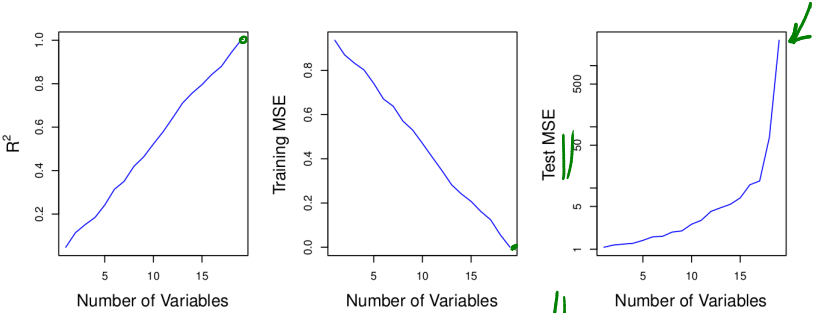
$p = 1$

$p = 1$

BUT

Given other sampled data Test MSE big

More issues with least squares on big p



- $n = 20$
- Regression on $p = 1, \cdots, 20$
- Y completely unrelated to variables

If only saw more variables = good!

Test MSE goes way up so more not better.

If n, p close R^2, AIC, BIC, C_p tend to do badly $\rightsquigarrow \hat{\sigma}^2$ bad

The answer to dealing with big p

Be less flexible!

Anything done so far for decreasing model flexibility helps:

Forward Step selection

Ridge regression / lasso

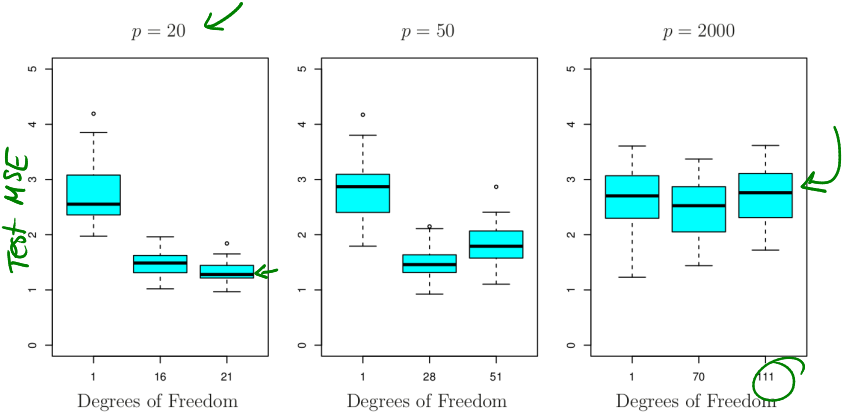
Subsets

PCR

PLS

Example with Lasso

$RSS + \lambda \sum |\beta_i|$



lower λ does better

high λ

- $n = 100$
- Boxplots = Test MSE
- DF = # non-zero coeffs

Key points

1. Regularization/shrinkage plays a key role in high dim problems

2. Appropriate tuning parameter selection is crucial

3. Test error tends to increase as dimensionality of problem (i.e. p) goes up UNLESS truly related to response

Curse of dimensionality

Phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings.

- You'd think more variables = better fitting model
- But really means opportunity for overfit

  Ex) previous, test MSE doubles from $p = 20 \to 2000$
- Adding noisy features → deterioration of model

Interpretation in high dimensions

Multi-collinearity: the concept that the variables in a regression might be correlated with each other

High dimensions: any variable in model can be written as linear combo of any other variable in the model

Can never know
- which are truly useful
- which are best for regression
- what "right" coeff should be

Any "good" model is one of many good model.

Reporting errors in high dimensions

- If $p > n$, easy to get "perfect" model w/ zero residuals

Using sum squared errors, p-vals, R^2, trad. measures of model fit as evidence of "good" fit is misleading

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Report results on independent test sets
CV errors

There are three kinds of lies: Lies, Damned Lies, and Statistics.

$\sim$ Mark Twain (among others), who attributed it to the British prime minister Benjamin Disraeli.

Next time

Next time:

- Ch 6.5 Lab

This week:

- Mon 3/14: 6.3 PCA and regression
- Weds 3/16: 6.4 High Dimensions
- Fri 3/18: 6.5 Lab
  - No homework!

Section 3

Go over midterm